FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Advanced)-2021

PAPER – 2

TEST DATE: 01-08-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total **54** questions.
- Each subject (PCM) has **18** questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section – B & Section-C.**

Section-A (01 – 06, 19 – 24, 37– 42) *this section contains* ***18 multiple*** *choice questions.*
Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:

Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
Negative Marks **: –2** In all other cases

Section-B (07 – 12, 25 – 30, 43– 48) *contains* ***18 Numerical*** *based questions with* ***Single digit integer*** *as answer, ranging from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-C (13 – 18, 31 – 36, 49– 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A coil of radius r(r << R) and mass m carries current i_2. Planes of two coils are mutually perpendicular and both the coils are free to rotate about common diameter.
Choose the correct statement/statements
   (A) The maximum Kinetic energy of smaller coil when both the coils are released, is
   $$K_{max} = \frac{\mu_0 \pi r^2 i_1 i_2 MR}{2(MR^2 + mr^2)}$$
   (B) Total energy of both the coils remains constant.
   (C) Total angular momentum of both the coils remains constant.
   (D) The potential energy of the system will be minimum when both the coils are in same plane and current in them are in same direction.

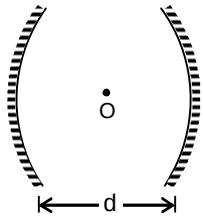
2. If the potential difference of Coolidge tube producing X-ray is increased, then choose the correct option(s).
   (A) the interval between $\lambda_{k\alpha}$ and $\lambda_{k\beta}$ increases
   (B) the interval between $\lambda_{k\alpha}$ and λ_0 increases
   (C) the interval between $\lambda_{k\beta}$ and λ_0 increases
   (D) λ_0 does not change
   Here λ_0 is cutoff wavelength and $\lambda_{k\alpha}$ and $\lambda_{k\beta}$ are wavelength of k_α and k_β characteristic X-rays.

3. A horizontal disc rotates freely about a vertical axis through its centre. A ring, having the same mass and radius as the disc, is now gently placed on the disc. After some time, they rotate with a common angular speed.
   (A) some friction exists between the disc and the ring.
   (B) the angular momentum of the disc plus ring is conserved.
   (C) the final common angular speed is $\frac{2}{3}$rd of the initial angular velocity of the disc.
   (D) $\frac{2}{3}$ rd of the initial kinetic energy changes to heat.

4. A point moves with deceleration along the circle of radius R so that at any time its tangential and normal acceleration are equal in magnitude. At the initial moment t =0, the velocity of the point is v_0. The velocity of the point will be
   (A) $v = \frac{v_0}{1 + \frac{v_0 t}{R}}$ at t second
   (B) $v = v_0 e^{-S/R}$ where s is path travelled on circle
   (C) $v = v_0 e^{-SR}$ where s is path travelled on circle

(D) $v = \dfrac{v_0}{1-\dfrac{v_0 t}{R}}$ at t second

5. The distance d between two identical concave mirrors of radius of curvature R. So, as to coincide the image I on the object O itself, which is placed at the mid point can be



(A) R
(B) 2R
(C) 3R
(D) R/2

6. The potential energy φ in joule of a particle of mass 1 kg moving in the x – y plane, obeys the law $\varphi = 3x + 4y$, where (x, y) are the coordinate of the particle in metre. If the particle is at rest at (6, 4) at time t = 0, then (only conservative force are acting on particle).
(A) the particle has constant acceleration
(B) the work done by the conservative forces from position of rest of the particle to the instant of the particle crossing x axis is 25 J
(C) the speed of the a particle when it crosses the y axis is 10 ms^{-1}
(D) the coordinates of the particle at time t = 4s are (–18, –28)

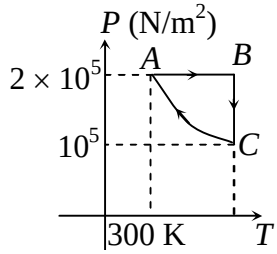
SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

7. A machine is blowing spherical soap bubbles of different radii filled with helium gas. It is found that if the bubbles have a radius smaller than 1 cm, then they sink to the floor in still air. Larger bubbles float in the air. Assume that the thickness of the soap film in all bubbles is uniform and equal. Assume that the density of soap solution is same as that of water (=1000 kg m^{-3}). The density of helium inside the bubbles and air are 0.18 kg m^{-3} and 1.23 kg m^{-3}, respectively. If d is the thickness (in μm) of the soap film of the bubbles which are just floating, find the value of 2d.

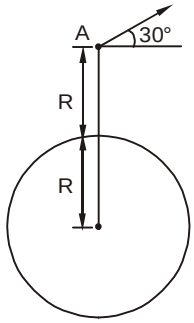
8. Find the mass ratio M_D/M_H of deuterium and hydrogen (in near integer value) if their $H\alpha$ lines have wavelengths of 6561.01 Å and 6562 Å respectively.

9. Two moles of a monatomic ideal gas is taken through a cyclic process shown on pressure(P)-temperature(T) diagram in figure. Process CA is represented as PT = Constant. If efficiency of given cyclic process is $1-\dfrac{3x}{12\ln 2+15}$, then find the value of 'x'.

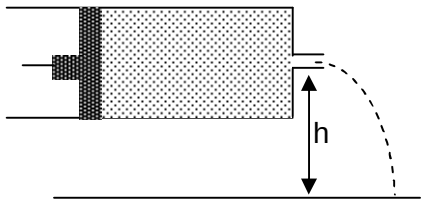


10. A stone is projected with initial velocity $u = 2\sqrt{2}\,m/s$ at an angle α = 45^0 with the horizontal, find the angular velocity in rad/sec of the stone with respect to the point of projection, when it is at its maximum height. (take g = 10 m/s^2)

11. A particle of mass m is fired from the point A (as shown in the figure) situated at a distance of 2R from the center of the planet with a velocity $\frac{1}{\sqrt{2}}$ times the escape velocity from point A. The radius of the planet is R. The maximum distance of the particle from the centre in the subsequent motion is nR. Find the value of n.



12. A tube has two area of cross-sections as shown in figure. The diameter of the tube are 8mm and 2mm. Find range (in m) of water falling on horizontal surface, if piston is moving with a constant velocity of 0.25 m/s, h = 1.25m
($g = 10\ m/s^2$)

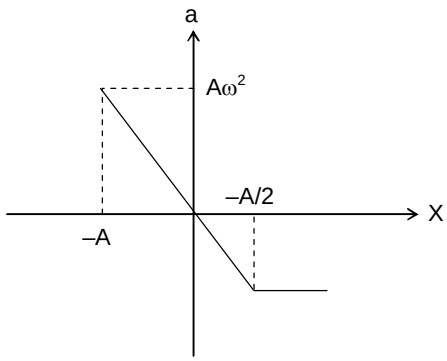


SECTION – C
(Numerical Answer Type)

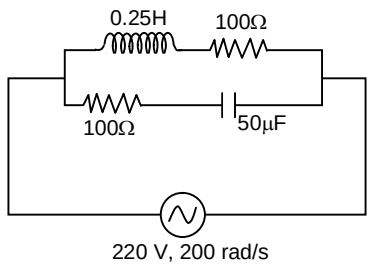
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A battery is made by joining identical cells in m rows each row having m cells. The current flowing in an external resistance R is i_1. Now if the number of rows 'm' and number of cells in each row 'n' are interchanged, the current in external resistance is i_2. If $\frac{m}{2} = 2$ and $\frac{R}{r} = 4$ then the value of $\frac{i_2}{i_1}$ is

14. Following is the acceleration vs position graph of a particle performing oscillatory motion along x-axis. ω = 2 rad/s. What is the time period of the given motion? (Take $\sqrt{3}$ = 1.73 and π = 3.00)



15. In the given Ac circuit, what is the power developed in watts?



16. A monkey of mass m is sitting on a platform of mass M. Monkey can jump with a velocity of 5 m/s making an angle 37^0 with the horizontal with respect to platform. To jump the monkey 1 meter with respect to the ground. Find out the value of m/M.

17. At a distance of 8 m from a point source, the loudness is 30 dB. What is the minimum distance in meter rounded value upto 2 decimal places the sound is inaudible. [Take $\sqrt{10}$ = 3.1623]

18. A parallel plate capacitor having plate area A and distance between the plates d, is filled with a medium of permitivity as $\in = \in_0 + \beta x$, when $0 < x < \frac{d}{2}$ and $\in = \in_0 + \beta(d - x)$, when $\frac{d}{2} < x < d$ (where x is taken along the plate separation). If $\frac{\beta d}{\in_0} = 2(e - 1)$, the value of equivalent dielectric constant is

Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. For reaction

$A(g) \rightleftharpoons B(g) + C(g) + D(g)$, equilibrium constant $= K_{C_1}$

$B(g) + D(g) \rightleftharpoons C(g) + E(g)$, equilibrium constant $= K_{C_2}$

1 mole of A(g) is placed in closed container at temperature T K and at one atmospheric pressure. A is heated, it is observed that total pressure at equilibrium is twice the partial pressure of A and pressure of gas C is observed to be 3/8 atm
Select the correct option

(A) Pressure of Gas E is 1/8 atm

(B) $K_{C_1} = \frac{1}{128}$

(C) $K_{C_2} = 3$

(D) $A \rightleftharpoons 2C + E$, $K_c = 3/128$

20. Which of the following is not a path function?
(A) entropy
(B) work done
(C) enthalpy at constant pressure
(D) free energy

21. Which of the following statements is/are correct?
(A) The heat of neutralization of a strong acid with a strong base is always the same
(B) The enthalpy of combustion is always negative
(C) A spontaneous change involves a lowering of free energy
(D) The enthalpy of an element in the standard state is assumed to be unity at 298 K.

22. What transition in the hydrogen spectrum would not have the same wavelength as the Balmer transition, n = 4 to n = 2 in the He^+ spectrum?
(A) n = 4 to n = 2
(B) n = 3 to n = 2
(C) n = 3 to n = 1
(D) n = 2 to n = 1

23. The bond order in O_2^+ is not same as in
(A) N_2^+
(B) N_2^{2-}
(C) He_2
(D) O_2^-

24. When a solid melts there will be
(A) an increase in enthalpy
(B) a decrease in free energy
(C) no change in enthalpy
(D) a decrease in internal energy

SECTION – B
(Single Digit Integer Type)

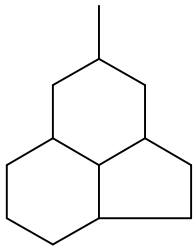
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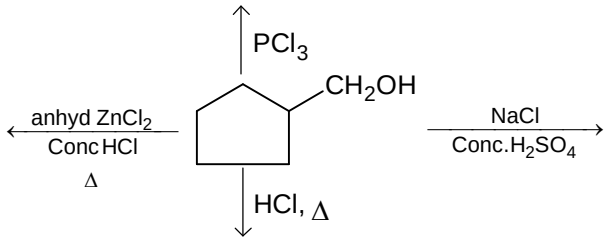
25. Compound x dissociates according to the reaction $2X(g) \rightleftharpoons 2Y(g) + Z(g)$, with degree of dissociation which is small compared to unity, if the expression for α in terms of equilibrium constant K_P and total pressure P is given as $\alpha = \left(\frac{2K_P}{P}\right)^{1/2}$. The value of x is

26. The number types of monochloroalkanes formed by chlorination of isobutane is

27. In alkaline medium, how many moles of potassium permanganate are required for complete oxidation of 3.5 moles of KI to KIO_3?

28. How many geometrical isomers are possible for the following complex?
$\left[Pt(NH_3)(NH_2OH)(NO_2)(py)\right]NO_2$

29. 
Total how many chiral C atoms are present in this compound?

30. 

In how many above reaction ring expansion takes place.

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. The molecular formula of Glauber's salt is $Na_2SO_4.10H_2O$. How much mass of oxygen is present in 32.2 g of the hydrated salt?

32. What is the standard free energy change of the following electrochemical reaction in kJ mol^{-1} unit? $X(s) + Y^{3+}(aq) \rightleftharpoons X^{3+}(aq) + Y(s); E^0_{Cell} = -0.8V$

33. The solubility product (K_{sp}) of $MgSO_4$ at a particular temperature is 4 x 10^{-4} mol^2 L^{-2}. What is the molarity of the saturated solution of $MgSO_4$?

34. Reaction of borazine with HCl forms a tri-chloro product(X). What is the molar mass of 'X' in g unit? [At. Mass of boron = 11]

35. The volume of 0.3 M ferrous ammonium sulphate solution required for the completion of redox titration with 20 mL of 0.1 M $K_2Cr_2O_7$ solution is - - - - - L.

36. The ionization energy of hydrogen atom is 13.6 eV and the first ionization energy of sodium atom is 5.1 eV. The effective nuclear charge experienced by the valence electron of sodium atom is (Round off to two decimal places)

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. The solution/s of the equation $9\cos^{12} x + \cos^2 2x + 1 = 6\cos^6 x \cos 2x + 6\cos^6 x - 2\cos 2x$ is/are

(A) $x = n\pi + \frac{\pi}{2}, n \in I$

(B) $x = n\pi \pm \cos^{-1}\sqrt[4]{\frac{2}{3}}, n \in I$

(C) $x = n\pi \pm \cos^{-1}\sqrt{\frac{2}{3}}, n \in I$

(D) $x = n\pi, n \in I$

38. If a straight line through a point $P(\alpha, 2), \alpha \neq 0$, meets the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ at A and D and x – and y–axes at B and C respectively such that $PA \times PD = PB \times PC$, then '$\alpha$' can lie in the interval

(A) $[7, \infty)$

(B) $(-12, -8)$

(C) $(-5, 0)$

(D) $(10, \infty)$

39. Let $f(x) = \cos^{-1}(2x^2 - 1)$ then:

(A) $f(x)$ is continuous in $[-1, 1]$

(B) $f(x)$ is derivable in (–1, 1)

(C) range of $f(x)$ is $[0, \pi]$

(D) derivative of $f(x)$ w.r.t. $\sin^{-1} x$ at $x = \frac{1}{2}$ is 2

40. A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx}(\sin x) - y\cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \to 0$ as $x \to \infty$ then the statement which is correct is:

(A) $\lim_{x \to 0} f(x) = 1$

(B) $\int_0^{\pi/2} f(x)\,dx$ is less than $\frac{\pi}{2}$

(C) $\int_0^{\pi/2} f(x)\,dx$ is greater than unity

(D) $f(x)$ is an odd function

41. Tangent drawn at point P(1, 3) of a parabola intersects its tangent at vertex at M(–1, 5) and cuts the axis of parabola at T. If R(–5, 5) is a point on SP; where S is focus of the parabola, then

(A) slope of axis is –3

(B) radius of circumcircle of ΔSMP is $\sqrt{\frac{5}{2}}$ units

(C) $(ST)^2 - (SM)^2 = (PM)^2$

(D) tangent cuts the axis of parabola at T (–3, 7)

42. Suppose that f(x) is a differentiable invertible function, $f(x) \neq 0$ and $h(x) = \int_1^x f(t)dt$. Given that $f(1) = f'(1) = 1$ and g(x) is inverse of f(x).,
Let $G(x) = x^2g(x) - xh(g(x)) \ \forall x \in R$. Which of the following are correct:

(A) G'(1) = 2

(B) G'(1) = 3

(C) G"(1) = 2

(D) G"(1) = 3

SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

43. Let $A = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$ and $x = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$. If $P = AXB$, $Q = BX^TA$ and $Tr.\left((PQ)^{10}\right) = a^{10} + b^{10}$, where $a < b$, then find the value of $b - 2a$ _____

(Tr (M) denotes trace of Matrix M and M^T denotes transpose of matrix M)

44. The straight line $\frac{x}{4} + \frac{y}{3} = 1$ intersects the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at two points *A* and *B*, there is a point *P* on this ellipse such that the area of ΔPAB is equal to $6(\sqrt{2} - 1)$, the number of such points (*P*) is

45. The roots of the equation $z^4 + az^3 + (12 + 9i)z^2 + bz = 0$ are the vertices of a square then area of square is _____

46. Let $\beta = \int_0^1 \frac{2 + 3x + 4x^2}{2\sqrt{x^2 + x + 1}}$, then find the value of β^2.

47. Water is drained from a vertical cylindrical tank by opening a valve at the base of the tank. It is known that the rate at which the water level drops is proportional to the square root of water depth y, where the constant of proportionality k > 0 depends on the acceleration due to gravity and the geometry of the hole. If time to empty the tank be T minutes, then $\frac{T}{10} =$

48. Suppose that the circle $x^2 + y^2 - 4ax - 2ay + 4a^2 = 0 (a > 0)$ touches the lines $y = mx$ at A and B such that AB = 3. Then the value of $\frac{4a}{\sqrt{5}}$ is

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. If $\sum_{k=0}^{100} \left(\frac{k}{k+1}\right) {}^{100}C_k = \frac{a\left(2^{100}\right)+b}{c}$ where $a,b,c \in N$, then find the least value of $\frac{(a+b+c)}{670}$.

50. If the sum of an infinite geometric series, whose first term is the limit of the function $f(x) = \left(\frac{9}{1-x^9} - \frac{7}{1-x^7}\right)$ as $x \to 1$ and the common ratio is the limit of function $g(x) = \frac{\tan x - \sin x}{\ln^3(1+x)}$ as $x \to 0$, is S then $\frac{7S}{100}$ is

51. A straight line cuts the x-axis at point $A(1,0)$, and y-axis at point B, such that $\angle OAB = \alpha \left(\alpha > \frac{\pi}{4}\right)$. C is middle point of AB and B' is a mirror image of point B with respect to line OC and C' is a mirror image of point C with respect to line BB', then the ratio of the areas of triangles ABB' and BB'C' is equal to

52. If number of points common between ||x| - |y|| = 1 and 2a |x| |y| + 1 = 2 |x| + a|y| are 8 then value of 3a (where a < 1) is ____.

53. $\vec{a}, \vec{b}$ and $\vec{c}$ be three non – coplanar vectors, and $\vec{d}$ be a non zero vector, which is perpendicular to $(\vec{a}+\vec{b}+\vec{c})$. Now if $\vec{d} = \sin x(\vec{a}\times\vec{b}) + \cos y(\vec{b}\times\vec{c}) + 2(\vec{c}\times\vec{a})$, then minimum value of $x^2 + y^2$ is equal to $\frac{\lambda\pi^2}{4}$, where λ is

54. Let 'ℓ' be a tangent line at the point $T\left(t, \frac{t^2}{2}\right)(t > 0)$ on the curve $C: y = \frac{x^2}{2}$. A circle S touches the curve C and the x – axis. Denote by $(x(t), y(t))$ the center of circle S. If $\lim_{r \to 0^+} \int_r^1 x(t)\, y(t)\, dt = \frac{\tan\frac{3\pi}{8}}{6a}$, then the value of a is